

JOONWON LEE

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SUMMARY

Statistics Ph.D. candidate and data scientist with hands-on property-insurance pricing experience and a rigorous background in predictive modeling, hypothesis testing, and statistical diagnostics. Built an end-to-end GLM/LightGBM workflow across 2.46M+ property-risk records and reusable Python/PyTorch pipelines for large, noisy datasets. Translates model evidence, uncertainty, and diagnostics into clear recommendations for business stakeholders.

EDUCATION

Ph.D. in Statistics, Rutgers University, Piscataway, NJ

Expected May 2027 | GPA: 3.8/4.0

M.S. in Statistics, University of Minnesota, Minneapolis, MN

Sep 2021 | GPA: 3.7/4.0

TECHNICAL SKILLS

Programming and Platforms: Python (NumPy, Pandas, SciPy, PyTorch, LightGBM, scikit-learn), SQL, R, Linux, Git, AWS EC2, HPC/SLURM; reproducible and restartable analytical pipelines

Predictive Modeling and Validation: Data preparation, feature processing, regression, GLM/GBM, supervised learning, model selection, calibration and validation, performance evaluation, feature attribution, hypothesis testing, Monte Carlo simulation, statistical diagnostics, documentation, and stakeholder communication

EXPERIENCE

JPMorgan Chase - Summer Quantitative Analytics Associate, Model Risk Governance and Review

Summer 2026

- Developed and validated a Sequential Probability Ratio Test for binary risk indicators, translating a monitoring requirement into explicit hypotheses, likelihood-ratio boundaries, and controlled stopping decisions.
- Combined Brownian-motion approximation, dynamic-programming probability propagation, and simulation to evaluate Type I/II error, boundary-crossing probabilities, stopping behavior, uncertainty, and model limitations.

Travelers - Data Science Leadership Program, Business Insurance Pricing Team

Summer 2025

- Developed an end-to-end LightGBM property-pricing pipeline across 2.46M+ records, including data preparation, feature processing, GLM benchmarking, model selection, validation, and performance diagnostics on AWS EC2.
- Evaluated predictive performance and feature-attribution patterns and presented pricing and risk-segmentation recommendations to business stakeholders.

RESEARCH AND ANALYTICAL SYSTEMS

Ph.D. Dissertation Research - Scalable Model Development and Statistical Diagnostics

Sep 2024 - Present

- **Scalable Modeling:** Developed an advection-aware Vecchia likelihood approximation for nonseparable Gaussian processes, replacing dense covariance operations with ordered local conditioning to fit up to 145,008 observations per day.
- **Data Quality and Root-Cause Analysis:** Built workflows for quality filtering, missingness tracking, time-specific coordinate offsets, nearest-neighbor matching, and irregular-to-regular grid mapping to separate systematic acquisition artifacts from random noise and model misspecification.
- **Model Diagnostics:** Designed scale- and frequency-resolved tests to identify whether discrepancies arise from low-frequency structure, high-frequency behavior, noise, missingness, or dependence; implemented reusable Python/PyTorch workflows for estimation, simulation, validation, and HPC execution.

ADDITIONAL

Coursework: Machine Learning, Probability Theory, Stochastic Processes, Statistical Computing, Advanced Theory of Statistics I-II, Data Structures and Algorithms, Linear Algebra, Microeconomics, Econometrics

Languages: Korean (Native); English (Fluent); Chinese (Basic)